

ESP32 SMART TEMPERATURE CONTROLLER WITH IOT MONITORING AND WEB DASHBOARD

Abstract

This project presents the design and implementation of an ESP32-based Smart Temperature Controller capable of automatically regulating water temperature using a DS18B20 digital temperature sensor, a relay-controlled immersion heater, and IoT-based monitoring. The system continuously measures water temperature and controls a 500W immersion heater through a relay module. The project incorporates safety features such as hysteresis control, emergency shutdown, sensor fault detection, and overtemperature protection. Real-time monitoring and control are achieved using both the Blynk IoT platform and a custom ESP32-hosted web dashboard.

Introduction

Temperature control systems are widely used in industrial automation, laboratories, water heating systems, food processing, and environmental monitoring applications. Conventional temperature controllers often require manual supervision and lack remote monitoring capabilities.

The objective of this project is to develop a low-cost, IoT-enabled temperature control system that automatically maintains water temperature within a desired range while providing real-time monitoring through a smartphone and web browser.

The ESP32 microcontroller was selected due to its built-in WiFi capabilities, processing power, and suitability for IoT applications.

Objectives

The primary objectives of this project are:

- To measure water temperature using a DS18B20 digital temperature sensor.
- To automatically control a heating element based on temperature thresholds.
- To implement relay-based heater switching.
- To provide remote monitoring using Blynk IoT.
- To develop a local web dashboard for real-time monitoring and control.
- To implement safety mechanisms including emergency shutdown and sensor fault protection.

Hardware Components

Component	Quantity
ESP32 DevKit V1	1
DS18B20 Waterproof Temperature Sensor	1
2N2222A NPN Transistor	1
1 k Ω Resistor	1
4.7 k Ω Resistor	1
5V Relay Module	1
500W Immersion Heater	1
Breadboard	1
Jumper Wires	Multiple
Power Supply	1

Software Tools

- Arduino IDE
- ESP32 Board Package
- Blynk IoT Platform
- HTML
- CSS
- JavaScript
- GitHub

System Architecture

The system consists of four major subsystems:

1. Temperature Sensing Subsystem
2. Processing and Control Subsystem
3. Heater Switching Subsystem
4. Monitoring and User Interface Subsystem

System Flow:

DS18B20 Sensor → ESP32 → 2N2222 Driver → Relay → Heater

Additionally:

ESP32 → WiFi → Blynk Dashboard

ESP32 → WiFi → Local Web Dashboard

Working Principle

The DS18B20 continuously measures the water temperature and sends digital temperature data to the ESP32 using the OneWire communication protocol.

The ESP32 compares the measured temperature with predefined temperature limits:

- Heater ON below 40°C
- Heater OFF above 45°C

This hysteresis prevents rapid relay switching.

The ESP32 controls the relay through a 2N2222 transistor driver circuit. The relay switches the AC power supplied to the immersion heater.

The system simultaneously updates temperature and heater status on:

- Blynk Mobile Dashboard
- Custom Web Dashboard

Circuit Description

DS18B20 Connections:

DS18B20 Pin	ESP32 Pin
VDD	3.3V
GND	GND
DQ	GPIO15

A 4.7 kΩ pull-up resistor is connected between DQ and 3.3V.

Relay Driver Connections

Component	Connection
GPIO26	1 kΩ Resistor
Resistor Output	2N2222 Base
2N2222 Emitter	GND
2N2222 Collector	Relay IN
Relay VCC	5V
Relay GND	GND

Heater Connections

Connection	Description
Live	Relay COM
Relay NO	Heater Terminal
Neutral	Heater Terminal

The relay switches only the live conductor while neutral remains directly connected to the heater.

Software Implementation

The software was developed in Arduino IDE using C++.

Major functions include:

- Temperature acquisition
- Hysteresis control
- Relay control
- Sensor fault detection
- Emergency shutdown
- Blynk communication
- Web server hosting

Safety Features

Sensor Failure Protection

If the DS18B20 becomes disconnected:

- Heater is immediately turned OFF.

Overtemperature Protection

If temperature exceeds 70°C:

- Heater is disabled automatically.

Emergency Stop

Users can manually disable the heater through:

- Blynk Dashboard
- Web Dashboard

IoT Integration

Blynk Dashboard

The Blynk interface provides:

- Live Temperature Display
- Heater Status Indicator
- Adjustable Setpoint
- Emergency Stop Control

Web Dashboard

A local ESP32-hosted dashboard was developed using HTML, CSS, and JavaScript.

Features include:

- Live Temperature Monitoring
- Heater Status Display
- Setpoint Adjustment
- Emergency Stop Function
- Mobile-Friendly Interface

Challenges Faced

ESP32 Upload Issues

Initially, code uploads failed due to antivirus interference and ESP32 bootloader connection issues.

Solution:

- Antivirus configuration was adjusted.
- Correct upload procedure was followed.

Relay Compatibility Issue

The relay module did not respond correctly when directly connected to ESP32 GPIO pins.

Cause:

- Logic level mismatch between ESP32 (3.3V) and relay module requirements.

Solution:

- A 2N2222 transistor driver stage was implemented.

Temperature Overshoot

The 500W immersion heater rapidly increased temperature in a small water volume (100–150 ml).

Solution:

- Hysteresis control was implemented.
- Improved sensor placement was used.
- Larger water volume was recommended.

Results

The developed system successfully:

- Measured water temperature accurately.
- Automatically controlled the heater.
- Maintained temperature within desired limits.
- Provided remote monitoring via Blynk.
- Hosted a local web dashboard.
- Implemented safety protections.

Experimental observations confirmed stable operation and reliable relay switching.

Future Improvements

Future enhancements may include:

- OLED Display Integration
- Data Logging
- SD Card Storage
- MQTT Communication
- Node-RED Dashboard
- Grafana Analytics
- PID Temperature Control
- Mobile Notifications
- Cloud-Based Monitoring

Conclusion

An ESP32-based Smart Temperature Controller was successfully designed and implemented. The system combines sensor interfacing, relay control, IoT communication, and web technologies to provide a complete temperature automation solution.

The project demonstrates practical skills in embedded systems, IoT development, sensor integration, hardware debugging, and web-based monitoring. The final system is reliable, scalable, and suitable for further industrial and academic applications.

GitHub Repository

ESP32-Smart-Temperature-Controller

<https://github.com/himanshu-rabha/ESP32-Smart-Temperature-Controller.git>

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